

Milestone 1 — Project Proposal

AML5005L / AIMAL5005L — Advanced Machine Learning

1. Project Title

Explainable Machine Learning for Healthcare Accessibility Assessment and Underserved-District Identification in India

2. Team Members

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3. Problem Statement

Healthcare access in India is not the same everywhere. Two districts can have the same number of hospitals but very different populations, so the actual accessibility is nowhere close to equal — 10 hospitals for 1 million people is not the same as 10 hospitals for 200,000 people. Data on healthcare infrastructure, services, population and geography already exists publicly, but it is scattered across different sources and not combined anywhere. We want to bring these sources together at the district level to see how accessible healthcare actually is, and to predict which districts are likely to slip into an underserved category next.

4. Motivation

- Healthcare infrastructure and staff are spread very unevenly across districts.
- Raw facility counts are misleading — they need to be normalized by population to mean anything.
- Even where a facility exists, distance can still make it hard for people to reach.
- Knowing which districts might become underserved is more useful than only knowing which ones already are.
- A risk score alone isn't very actionable — knowing why a district is flagged matters more for planning.

5. ML Task

We're splitting this into two connected pieces:

- Building an index: combine normalized availability, capacity and service-utilization indicators into a District Healthcare Accessibility Index (D-HAI) for each district-year.
- Prediction: a binary classification task — given a district's indicators in year t , predict if it will be underserved in year $t+1$ ($X_t \rightarrow Y_{t+1}$).

We also want to explain the predictions, not just output a label, so SHAP is part of the core plan rather than an extra we add later if time permits.

6. Proposed Dataset

We're building our own district-year dataset by combining a few public sources (no Kaggle):

- Rural Health Statistics (RHS) reports — facility counts, beds, doctors, nurses, health workers.
- National Health Profile / data.gov.in — service utilization and public health indicators (used instead of the login-gated HMIS dashboards, which aren't reliably accessible).
- Census of India / population projections — for per-capita normalization.

- LGD (Local Government Directory) codes — as the common district identifier across sources and years.
- Publicly available facility location data, where we can get it — for a rough geographic-accessibility measure like facility density or nearest-facility distance.

7. Data Collection Methodology

We'll pull raw reports/tables directly from the official sources above for the most recent years we can get consistently, then:

- Match districts across sources using LGD codes, not district names (names aren't consistent across sources or years).
- Clean and line everything up into one district-year table.
- Mark missing values as missing instead of quietly filling them with zero.
- Build per-capita / per-area features instead of using raw counts directly.
- Define the underserved label using a fixed threshold on D-HAI, applied the same way across all years.

We're keeping our scripts and a source list so the dataset build can be reproduced.

8. Expected Number of Observations

India has around 750 districts (this shifts slightly year to year as new districts get created). If we can get 3–5 years of usable data across sources, that's roughly 2,250–3,750 district-year rows. The real number will come down once we see how much overlap there actually is across sources after cleaning — some years/districts will likely drop out due to missing data, so we expect the final count to be somewhat lower than this.

9. Candidate ML Algorithms

- Logistic Regression — baseline, easy to interpret.
- Decision Tree — simple non-linear baseline.
- Random Forest — stronger baseline, gives feature importance for free.
- SVM — for comparison.
- Gradient Boosting / XGBoost / LightGBM — likely candidates for the final model.

SHAP will be applied on whichever model performs best, for both overall and per-district explanations.

10. Expected Outcome

- A self-built, documented district-year healthcare dataset for India.
- The D-HAI index and district rankings based on it.
- A classifier that predicts next-year underserved status, validated on time (not a random split).
- SHAP explanations showing what's driving underserved-risk predictions, overall and per district.
- District-level maps/visuals showing accessibility and risk patterns.

Note: We're scoping this to what's realistically doable with public data in the course timeline. Things like affordability and acceptability of care aren't covered by our data, so we're not claiming to measure those — this is an accessibility proxy/index, not a complete accessibility measure.